Artificial Intelligence and Credit Risk Management in Chinese State Banking: The Case of ICBC

Author: Patrycja Oliwia Pyzikiewicz

Executive Summary

This report analyses non-performing loan trends at the Industrial and Commercial Bank of China (ICBC) and proposes an AI-driven credit risk assessment model using Gradient Boosting Machine (XGBoost). Financial ratio analysis (2012–2023) shows rising loan-to-deposit ratios, declining revenue growth, and increasing cost-to-income ratios – indicating inefficiencies in lending decisions. The proposed GBM model aims to reduce false approvals and false rejections, improve risk management, and lower NPL rates. The report also addresses bias, explainability (SHAP), and implementation costs. While AI offers significant efficiency gains, human oversight remains essential.

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1. Introduction

The Industrial and Commercial Bank of China (ICBC) is one of the largest commercial banks globally, ranking among the top in terms of tier-one capital and total assets (World Economic Forum, n.d.). Despite its strong market position, ICBC faces significant challenges, particularly the rising amount of non-performing loans (NPLs), a broader issue that has affected Chinese banks in general. According to a study by Zhang, Wang, and Xie (2024), the total NPLs in the Chinese banking sector have surged from $60 billion in 2011 to $421 billion in 2022. This growing issue, which is made worse by monetary policy frictions that make it more difficult for central bank policies to be effectively transmitted, weakens banks' capacity to control credit risk and maintain profitability. For ICBC, achieving its strategic objectives of financial stability and operational efficiency requires addressing inefficiencies in risk management and loan origination. This report evaluates ICBC's financial performance through key ratio analysis and examines how AI can enhance operational efficiency, improve credit risk assessment, and explains bias and ethical issues of the use of AI.

2. Operational Inefficiencies Based on Financial Ratio Analysis

An analysis of ICBC's financial ratios from 2012 to 2023, obtained from the bank's provided financial statements, highlights several inefficiencies affecting its performance. These ratios were selected for their ability to provide insights into ICBC's operational efficiency, profitability, and risk management, offering a comprehensive understanding of its financial health over time.

Year	2023	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012
Loan to Deposit (LDR)	6.27	6.57	6.48	5.11	4.91	4.34	3.84	3.81	3.81	3.06	2.94	2.7
Non-performing Loan (NPL)	0.01	0.01	0.01	0.02	0.01	0.02	0.02	0.02	0.02	0.01	0.01	0.01
Revenue Growth Rate	-3.8	-7.3	6.64	3.34	10.9	6.76	7.46	-3.1	5.59	11.6	9.3	-
Return on Assets (ROA)	0.43	0.41	0.37	0.36	0.37	0.39	0.4	0.42	0.4	0.42	0.45	0.44
Return on Equity (ROE)	0.1	0.11	0.11	0.11	0.12	0.13	0.13	0.14	0.15	0.18	0.21	0.21
Cost to income Ratio (CTIR)	1.02	1.08	1.23	1.26	1.2	1.1	1.01	0.88	0.95	0.84	0.75	0.75

ICBC Financial Ratio Analysis (2025)

The Loan-to-Deposit Ratio (LDR) has risen from 2.70 in 2012 to 6.27 in 2023, indicating a shift toward more aggressive lending. While this may enhance profitability, it also raises concerns regarding liquidity management if lending exceeds deposit growth (Eltweri et al., 2024).

ICBC's Revenue Growth Rate has been inconsistent, with notable declines in 2022 (-7.3%) and 2023 (-3.8%). This trend raises concerns about the bank's ability to sustain income growth, which is essential for long-term stability (Twum, Agyemang and Sare, 2022).

Additionally, Return on Assets (ROA) has remained relatively low, at 0.43 in 2023, indicating that ICBC's profitability on total assets has not improved over time. However, according to Einstein (2025) banks typically have lower ROAs due to their assets generating lower profit margins.

A critical concern is the rising Cost-to-Income Ratio (CTIR), which increased from 0.75 in 2012 to 1.02 in 2023. This indicates that ICBC's operational costs have surpassed income, suggesting failures in expense management. Over time, such inefficiencies can erode the bank's competitive position, making it crucial to address cost management without sacrificing operational effectiveness (Turdibayeva, 2024).

Furthermore, Return on Equity (ROE) has declined from 0.21 in 2012 to 0.10 in 2023, reflecting reduced profitability for shareholders. ROE is a key measure of how efficiently shareholder capital is used to generate profit, making it a critical indicator for investors, regulators, and bank executives (Norman, 2017). A lower ROE suggests reduced asset profitability or lower leverage, both of which can impact the bank's ability to meet profitability targets and attract investment (Norman, 2017).

ICBC maintains an exceptionally low Non-Performing Loan (NPL) ratio of 0.01–0.02, which, while reflecting strong credit quality and effective risk management, may also indicate an overly cautious lending approach, where the bank might be prioritizing safety over aggressive growth. This conservative strategy, while reducing the risk of defaults, may limit the bank's profitability and growth potential in a more competitive and dynamic market environment (Mahyoub and Said, 2021).

AI-driven solutions can help address these inefficiencies by optimizing credit risk assessment, enhancing loan approval processes, and improving cost management, aligning with ICBC's broader strategic objectives for financial stability and growth.

3. Development and Deployment of AI

a) Rationale and Justification for AI/ML Integration

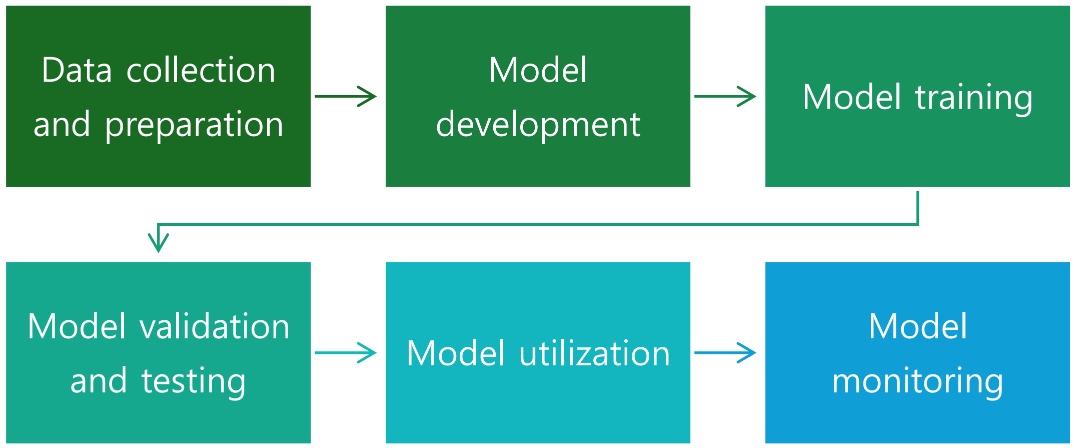
Currently, ICBC's reliance on traditional credit scoring models and manual decision-making has led to inefficiencies and increased risk – as its proven by the analysis provided previously (Icbc.com.cn, 2023). To improve ICBC's lending and risk management processes, the integration of AI-based credit risk assessment models is essential rates (Brown, 2024, Sadok and El Hadi El Maknouzi, 2025). One of the most effective approaches is the Gradient Boosting Machine (GBM), which is excellent at recognizing subtle patterns in large data sets and has demonstrated high accuracy in financial applications (Rahman and Alam, 2023).

GBM works as a supervised learning model, meaning that it is trained on historical data in which both input characteristics (such as an applicant's income, credit history, and repayment behavior) and their corresponding outputs (such as whether an applicant defaulted or repaid a loan) are known (Natekin and Knoll, 2013). During training, the model learns to map these input variables to a desired output, minimizing prediction errors. This supervised approach allows GBM to identify relationships between different factors that affect loan outcomes, allowing it to make highly accurate predictions when evaluating new applications (Rahman and Alam, 2023).

In supervised learning, the model iteratively improves its performance by minimizing a defined loss function (such as the root mean square error for regression or the log loss for classification), ensuring that future predictions become more accurate over time. For ICBC, this means that GBM can effectively predict the probability of default by analyzing patterns learned from previous applicants, ultimately helping the bank make more informed and data-driven lending decisions (Natekin and Knoll, 2013).

Using this supervised learning capability, GBM ensures that ICBC can reduce the number of errors in loan approvals — minimizing the chances of approving high-risk applicants (false positives) and rejecting low-risk borrowers (false negatives) (Rahman and Alam, 2023). This translates into better risk management, lower nonperforming loan (NPL) rates, and improved overall bank profitability.

b) Development and Testing Approach



Construction of GBM-model (2025)

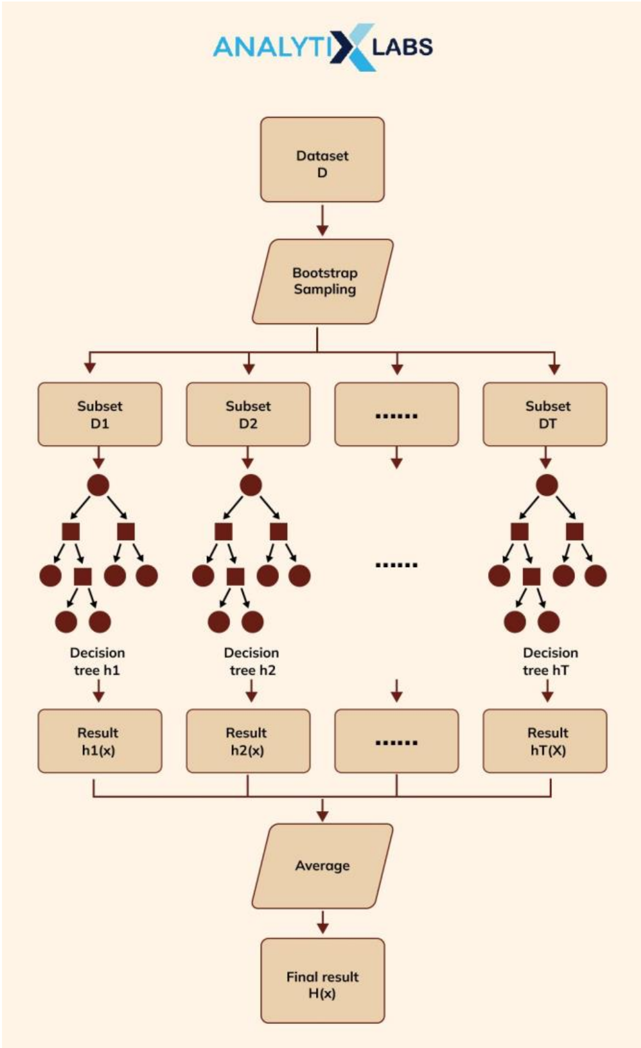
To successfully implement a GBM-based credit risk model (Uddin et al., 2023), a structured and technical development approach is proposed:

1. Data Collection and Preparation

ICBC will gather historical loan application data spanning from 2015 to 2024, capturing key details such as loan amounts, applicant income, employment history, repayment patterns, and relevant macroeconomic indicators. This data will be carefully cleaned to handle missing values, convert categorical information into numerical formats, and normalize continuous variables for consistency (Uddin et al., 2023). To enhance predictive accuracy, additional variables like the loan-to-income ratio, debt service ratio, and repayment frequency will be derived through feature engineering.

2. Model Development

The Gradient Boosting Machine (GBM), specifically using XGBoost, is ideal for this application due to its ability to handle large, complex financial datasets (Rahman and Alam, 2023).



Gradient Boosting Algorithm (Tripathi, 2022)

The GBM algorithm starts by creating a weak initial model and then adds new trees to correct prediction errors iteratively (Natekin and Knoll, 2013). Each tree focuses on the residuals (errors) from the previous tree, refining the model's accuracy with every iteration. This sequential process allows GBM to identify complex patterns in the data, making it highly effective for predicting loan outcomes.

To maintain transparency and explainability, SHAP (SHapley Additive exPlanations) is used to identify and rank the most influential variables affecting loan decisions, ensuring that ICBC's model remains aligned with regulatory requirements (Teuku Rizky Noviandy, Ghalieb Mutig Idroes and Hardi, 2024).

3. Model Training

The model will be trained on 75% of the dataset, leaving 25% for validation to ensure accurate performance assessment (Uddin et al., 2023). To prevent overfitting and maximize predictive power, cross-validation techniques will be applied during training. Hyperparameters such as the learning rate (0.05), maximum tree depth (6), number of estimators (500), and minimum child weight (1) will be fine-tuned for optimal performance (GeeksforGeeks, 2024).

4. Model Validation and Testing

Model performance will be evaluated using the remaining 25% of the dataset (Uddin et al., 2023). Accuracy will measure overall correctness, while precision and recall will assess the balance between identifying high-risk borrowers and minimizing false approvals. The AUC-ROC curve will determine how effectively the model differentiates between defaulters and non-defaulters.

5. Model Utilization

After successful validation, the model will be deployed via a real-time API integrated into ICBC's loan processing system. Routine applications will be processed automatically, with low-risk and high-risk cases receiving immediate decisions. Borderline applications, however, will be flagged for manual review by credit analysts to ensure that complex cases receive appropriate attention.

6. Model Monitoring

A 6-month pilot phase will process 10,000 loan applications to evaluate the model's real-world performance. Throughout this period, the model's predictions and outcomes will be closely monitored. Continuous evaluation will identify any areas for refinement, and retraining pipelines will be in place to keep the model responsive to evolving borrower behavior and market conditions.

c) Cost Analysis and Management

Implementing an AI/ML system at ICBC involves several costs, particularly in terms of IT infrastructure, staffing, and ongoing technical support (Walturn.com, 2025). If ICBC decides to build the system in-house, the bank will need to invest significantly in compute resources, cloud storage, and dedicated staff, including data scientists and engineers. This option provides greater flexibility and control over the model but requires ongoing effort in terms of maintaining and improving the model (Walturn.com, 2025).

Alternatively, purchasing an off-the-shelf AI solution can reduce implementation time and dependence on internal resources, but comes with limitations in terms of customization. These systems may not be a perfect fit for ICBC's specific needs, and there may be long-term dependence on external vendors (Walturn.com, 2025).

The introduction of AI will also change the way current credit analysts and loan officer's work. While automating routine tasks can reduce the need for manual intervention, it also opens up opportunities to retrain staff to handle more complex cases, increasing their role in the decision-making process (Muhammad Fahad et al., 2024). The three-month pilot phase, involving approximately 10,000 applications, will allow ICBC to evaluate the effectiveness of the model and make any necessary changes.

While the initial investment is significant, the long-term benefits—such as faster processing, greater accuracy, and improved risk management—offer significant returns to ICBC's growth and efficiency.

4. Bias and Ethical Issues in AI

Biases and ethical concerns in financial decision-making pose significant challenges for banks like ICBC. When AI-driven lending decisions rely on historical data, past inequalities in loan approvals can carry over into future decisions. If certain demographic groups were previously denied loans at higher rates, automated systems might reinforce these patterns, leading to unfair outcomes (Natekin and Knoll, 2013). This not only harms customers by denying credit to those who deserve it but also damages ICBC's reputation and regulatory relationships.

Operational efficiency may also suffer if AI-driven lending decisions are not properly monitored. If the system incorrectly classifies low-risk borrowers as high-risk, ICBC could lose potential customers and revenue. Conversely, if it grants loans to risky applicants, default rates may rise, threatening the bank's financial stability. Transparency is another key concern—when customers do not understand why they were denied a loan, it can lead to mistrust. Regulators and internal auditors also require clarity to assess fairness in lending decisions. However, AI models often function as "black boxes," making it difficult to explain why a particular decision was made, which can undermine trust in automated lending (Rahman and Alam, 2023).

Another major limitation of AI is its inability to handle rare, unpredictable events known as "black swan" events (Kolt, 2023). Economic crises, sudden market fluctuations, or global disruptions can render historical data unreliable, leading to flawed risk assessments. AI models trained on past data may fail to recognize these extreme scenarios, exposing ICBC to unexpected financial losses.

Common issues in automated lending include biased data, flawed decision-making patterns, and feedback loops that reinforce past mistakes. Data bias occurs when the system is trained on information that does not fully represent diverse borrowers. Feedback loop bias can deepen disparities by repeatedly disadvantaging historically underprivileged groups (Zou and Khern-am-nuai, 2022).

To address these risks, ICBC must implement safeguards, including improving data quality, regularly auditing lending patterns for bias, and adhering to strict regulatory guidelines. Human oversight should remain central, especially for complex cases. By combining technology with responsible management practices, ICBC can enhance efficiency while maintaining fairness, customer trust, and regulatory compliance.

5. Conclusion

Integrating AI-driven credit risk assessment at ICBC can enhance loan approvals, reduce non-performing loans, and improve risk management. Machine learning models like Gradient Boosting Machines (GBM) can analyze large datasets to make precise lending decisions. However, AI also presents challenges, such as biased data, opaque decision-making (black box problem), and vulnerability to unexpected financial crises (black swan events). Without proper oversight, AI may reinforce existing biases or fail to adapt to sudden economic shifts.

To mitigate these risks, ICBC must ensure transparency through explainability tools, regular audits, and human oversight for complex cases. Compliance with regulatory standards is essential to maintain trust and accountability. Additionally, continuous model updates will be necessary to adapt to changing market conditions. While AI can drive efficiency and profitability, its success depends on responsible implementation, fairness in decision-making, and a balance between automation and human expertise.

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